

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

V Semester of BSc Biotechnology/ Microbiology/ Biochemistry Examination March 2019

Course code & Name: BS501 Basic Molecular Biology II

14.03.2019

Thursday

10:00 AM To 10:20 AM

Maximum Marks: 10

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

Q - I Choose the correct answer for the following questions.

10

1. Genetic material in mammals is
(i) DNA (ii) RNA
(iii) Proteins (iv) Lipids
2. Select the right flow
(i) Gene→Protein→mRNA
(ii) Protein→Gene→mRNA
(iii) Gene→mRNA→Protein
(iv) All the above are wrong
3. Select the most appropriate association with rRNA
(i) Ribosomes (ii) Lysosome
(iii) Cytoskeleton (iv) Golgi Complex

4. Ribosomes are made up of
- | | |
|-----------------------|----------------------|
| (i) RNA and Lipid | (ii) DNA and Protein |
| (iii) RNA and Protein | (iv) DNA and Lipid |
5. Transcription leads to synthesis of
- | | |
|-----------|--------------------|
| (i) RNA | (ii) Protein |
| (iii) DNA | (iv) All the above |
6. The non-variable 3' terminal sequence found in tRNA is
- | | |
|-----------|----------|
| (i) CCG | (ii) CCU |
| (iii) CCC | (iv) CCA |
7. Which of the following does not possess DNA
- | | |
|------------------|------------------|
| (i) Mitochondria | (ii) Chloroplast |
| (iii) Nucleus | (iv) Ribosome |
8. Which of the following is not a stop codon
- | | |
|-----------|----------|
| (i) TAA | (ii) TAG |
| (iii) TGA | (iv) TTT |
9. Sigma factors are required by eukaryotes for transcription True/ False
10. All types of RNA leads to protein synthesis True/ False

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14.03.2019 Thursday 10:21 AM To 12 Noon Maximum Marks: 25

Instructions:

1. **Section I and II must be attempted in SINGLE ANSWER SHEET.**
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of non-programmable calculator is allowed.
4. Show necessary calculations.

SECTION -I

Q - II Answer the following questions as directed **10**

1. (i) Differentiate prokaryotic and eukaryotic transcription (5 points) 5
OR
(ii) Describe prokaryotic translation
2. Describe **ANY ONE** 5
(i) *Lac* operon
(ii) *Trp* operon

SECTION – II

Q-III Answer the following questions as directed **15**

1. (i) Describe active centers of Ribosomes 5
OR
(ii) What is c-value paradox? Discuss briefly
2. **Attempt ANY ONE** 4
(i) RNA processing in eukaryotes
(ii) Post translational modifications
3. Describe briefly **ANY THREE** 6
(i) DNA topology
(ii) Mobile DNA
(iii) Chromatin remodeling
(iv) Gene Cluster
(v) Genetic switches of phage lambda